Chloride

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Identity Sheet

Property	Value
Name	Chloride
Other names	Chloride ion, Cl^-
Chemical formula	Cl^-
Atomic/molecular weight	35.45 g/mol
CAS number	16887-00-6 (chloride ion)
PubChem CID	312
Appearance	Colourless in solution

Structural formula:

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Cl(-) (monovalent anion, fully hydrated in solution)
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Chloride is the **principal extracellular anion** in the body. It accompanies sodium in maintaining ECF volume, osmolality, and electrical neutrality. In urine, chloride excretion largely parallels sodium excretion and is clinically valuable in assessing volume status, metabolic alkalosis aetiology, and acid-base disorders.

Molecules Not to Be Confused With

Molecule	Formula	MW (g/mol)	Key Difference
Bicarbonate (HCO_3^-)	HCO_3^-	61.02	Second major ECF anion; inversely related to Cl^- in acid-base disorders
Bromide (Br^-)	Br^-	79.90	Halide analogue; can interfere with some Cl^- assays
Fluoride (F^-)	F^-	19.00	Another halide; different biological role (dental health)

Medical Information

Origin

Endogenous Chloride is not synthesised. Total body Cl^-: ~33 mmol/kg (~2,300 mmol in a 70 kg adult). Predominantly extracellular (~88% in ECF). Also in gastric HCl secretion (~150 mmol/L in gastric juice).

Exogenous Dietary NaCl is the primary source. Typical intake: 100–250 mmol/day Cl^- (matching Na^+). Found in processed foods, table salt, bread, dairy, and cured meats.

Primary & Secondary Biological Roles

Primary role: - **Electrical neutrality and osmolality:** Cl^- is the main counter-ion to Na^+ in ECF. - **Gastric acid production:** Parietal cells secrete HCl for digestion and antimicrobial defence.

Secondary roles: - **Acid-base balance:** Cl^- shifts (chloride-responsive vs -resistant alkalosis); renal Cl^- handling linked to H^+ and HCO_3^- transport. - **GABA receptor function:** Cl^- channels mediate inhibitory neurotransmission. - **Immune function:** Myeloperoxidase uses Cl^- to generate hypochlorous acid (HOCl) for microbial killing.

Catabolism and Elimination Pathway

- Cl^- is freely filtered (~18,000 mmol/day) and ~99% reabsorbed, primarily in the proximal tubule (paracellular), loop of Henle (NKCC2), and distal nephron.
- Urinary Cl^- excretion matches dietary intake at steady state.
- Other losses: sweat (~10–40 mmol/L), GI (vomiting: significant Cl^- loss; diarrhoea: variable).

Expression in Humans

Normal Levels

Compartment	Reference Range
Serum chloride	96–106 mmol/L
Urinary chloride (24-h)	110–250 mmol/day
Urinary chloride (spot)	20–200 mmol/L
Sweat chloride	<30 mmol/L normal; >60 mmol/L diagnostic for CF

Factors Influencing Levels **Increased urinary Cl:** - High NaCl diet - Diuretics (loop, thiazide) - Salt-wasting nephropathy - Bartter syndrome, Gitelman syndrome - Mineralocorticoid excess (with metabolic alkalosis) - SIADH

Decreased urinary Cl: - Vomiting (Cl^- loss in gastric fluid; renal conservation) - Volume depletion - Chloride-responsive metabolic alkalosis (UCl <20 mmol/L) - Heart failure, cirrhosis - Low-salt diet

Associated Pathologies

Condition	Urine Cl Pattern	Key Symptoms
Chloride-responsive alkalosis (vomiting, NG suction)	UCl <20 mmol/L	Metabolic alkalosis corrected by saline
Chloride-resistant alkalosis (hyperaldosteronism)	UCl >20 mmol/L	Hypertension, hypokalaemia
Cystic fibrosis	Sweat Cl >60 mmol/L (not urine)	Recurrent lung infections, pancreatic insufficiency
Bartter/Gitelman syndromes	High UCl despite hypokalaemia	Metabolic alkalosis, normotension

Presence in Urine

Should it be normally present? Yes — chloride is a normal major urinary electrolyte.

Normal urinary levels: 110–250 mmol/day (24-h); spot 20–200 mmol/L.

Form in urine: Cl^- ion — fully dissociated. No crystallisation.

Pathological significance:

Urinary Cl	Possible Causes	Prevalence
High	High salt diet, diuretics, salt-wasting	Common
Low (<20 mmol/L)	Vomiting, volume depletion, Cl-responsive alkalosis	Common

Solubility: Chloride salts (NaCl, KCl) are highly soluble. No crystalluria.

Detection in Urine

Available Clinical Assays

1. **Ion-selective electrode (ISE):**
   - **Principle:** Ag/AgCl-based or polymer membrane ISE generates Nernstian potential for Cl^-.
   - **Detection:** Potentiometric.
   - **LOD:** ~1 mmol/L. CV <2%.
   - **Advantages:** Standard on automated analysers; rapid.
2. **Coulometric-amperometric titration (Cotlove chloridometer):**
   - **Principle:** Cl^- reacts with Ag^+ generated coulometrically; endpoint detected amperometrically.
   - **Detection:** Current change at silver electrodes.
   - **LOD:** ~1 mmol/L.
   - **Advantages:** Reference method; very precise.
   - **Disadvantages:** Manual; time-consuming.
3. **Mercurimetric titration:**
   - **Principle:** $Hg(NO_3)_2$ titrant reacts with Cl^- to form soluble, un-ionised $HgCl_2$; endpoint by diphenylcarbazone indicator (blue-violet).
   - **Detection:** Visual endpoint or 515 nm photometric.
   - **LOD:** ~5 mmol/L.

Optimal Urine Type for Measurement

- **24-hour urine** for total excretion assessment.
- **Spot urine** for metabolic alkalosis workup (UCl <20 or >20 mmol/L determines Cl-responsive vs -resistant).
- No special preservatives. Stable at room temperature.

Actual Gold Standard

Coulometric-amperometric titration (chloridometer) is the reference method. ISE on automated platforms (Roche, Siemens, Abbott) is the universal clinical method.

Interferences in Measurement

Interference	Effect	Mechanism
Bromide	Positive bias (ISE)	Cross-reactivity of Cl^- ISE membrane
Iodide	Positive bias (ISE)	Similar cross-reactivity
High protein	Minor effect	Excluded volume (indirect ISE)
Thiocyanate	Positive bias	Halide-like behaviour

Research Detection Methods

Spectroscopy Detection (UV-Vis / NIR)

- Cl^- has **no UV-Vis absorption** in aqueous solution. No direct spectroscopic detection.
- **Indirect colorimetric:** Chloride displaces thiocyanate from $Hg(SCN)_2$; free SCN^- reacts with Fe^{3+} → red $Fe(SCN)_3$ complex (460–480 nm). LOD: ~0.5 mmol/L. Sample prep: reagent addition.
- **NIR:** Not applicable.

Fluorescence Detection

- Cl^- is non-fluorescent. Indirect methods:
- **Chloride-quenching probes:** Lucigenin (bis-N-methylacridinium, Ex 368 / Em 505 nm) and SPQ (6-methoxy-N-(3-sulfopropyl)quinolinium, Ex 344 / Em 443 nm) are quenched by Cl^- via collisional quenching. LOD: ~1 mmol/L.
- **Sample prep:** Add probe to urine or diluted sample; measure fluorescence decrease.

Raman Detection Cl^- (monatomic ion) has no Raman-active vibrational modes. Not detectable by Raman.

FTIR Detection Cl^- has no IR absorption in solution (monatomic). Not detectable by FTIR.

Voltammetry Detection

- **Potentiometric ISE** is the standard (see above).
- **Amperometric titration** (chloridometer) at silver electrodes.
- Direct voltammetric oxidation of Cl^- ($Cl^- \rightarrow Cl_2$) requires very high potentials (+1.36 V vs SHE) — not practical in urine.

Other Detection Technologies

1. **Ion chromatography (IC):** Anion exchange + conductivity detection. LOD: ~0.01 mmol/L. Multi-anion (Cl^-, SO_4^{2-}, NO_3^-, PO_4^{3-}). Reference for research.
2. **Capillary electrophoresis:** UV indirect detection. LOD: ~0.1 mmol/L.
3. **Paper-based $AgNO_3$ test:** $Ag^+ + Cl^- \rightarrow AgCl$ precipitate (white). Semi-quantitative. LOD: ~10 mmol/L.

Summary Table: Detection Methods Comparison

Method	LOD	Key Parameter	Sample Prep	Strengths
ISE	~1 mmol/L	Potentiometric	None	Standard, fast
Chloridometer	~1 mmol/L	Amperometric	Dilution	Reference, precise
Colorimetric (Hg/Fe)	~0.5 mmol/L	460-480 nm	Reagent	Simple, cheap
Lucigenin fluorescence	~1 mmol/L	Ex 368/Em 505 nm	Probe addition	Imaging capable
Ion chromatography	~0.01 mmol/L	Conductivity	Dilution	Multi-anion
Raman / FTIR	N/A	N/A	N/A	Not applicable

References

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